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MILITARY STANDARD

INSPECTION, LIQUID PENETRANT

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DEPARTMENT OF DEFENSE
Washington, DC 20402

Inspection, Liquid Penetrant

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2. Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: AFWAL/MLSE, Standardization Manager, Wright-Patterson AFB, Ohio 45433-6533, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1. SCOPE

1.1 Purpose. This standard establishes the minimum requirements for conducting liquid penetrant inspection of nonporous, metal and nonmetal components.

1.2 Application. The penetrant inspection processes described in this standard are applicable to in-process, final, and maintenance (in-service) inspections. These processes are applicable for the detection of discontinuities that are open or connected to the surface of the component under inspection.

1.3 Classification. Penetrant inspection processes are classified in accordance with the material classification contained in MIL-I-25135.

1.3.1 Penetrant systems. Penetrant systems covered by this standard shall be of the following types, methods and sensitivity levels.

Type

Type I - Fluorescent dye.
Type II - Visible dye.
Type III - Visible and fluorescent dye (dual mode).

Method

Method A - Water-washable.
Method B - Post emulsifiable, lipophilic.
Method C - Solvent removable.
Method D - Post emulsifiable, hydrophilic.

Sensitivity

Sensitivity Level 1 - Low.
Sensitivity Level 2 - Medium.
Sensitivity Level 3 - High.
Sensitivity Level 4 - Ultrahigh.

1.3.2 Developers. Developers shall be of the following forms:

Form a - Dry powder.
Form b - Water soluble.
Form c - Water suspendable.
Form d - Nonaqueous.
Form e - Specific application.

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1.3.3 Solvent removers. Solvent removers shall be of the following classes:

Class (1) - Halogenated.
Class (2) - Non-halogenated.
Class (3) - Specific application.

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2. REFERENCED DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation form a part of this standard to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-I-25135	Inspection Materials, Penetrant
MIL-P-47158	Penetrant Inspection, Soundness Requirements for Materials, Parts, and Weldments

STANDARDS

MILITARY

MIL-STD-410	Nondestructive Testing Personnel Qualification and Certification
MIL-STD-792	Identification Marking Requirements for Special Purpose Components
MIL-STD-45662	Calibration Systems Requirements

(Copies of specifications, standards, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this standard to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the solicitation. The issues of documents which have not been adopted shall be those in effect on the date of the cited DODISS.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D95	Water in Petroleum Products and Bituminous Materials by Distillation, Test for
ASTM D2512	Compatibility of Materials with Liquid Oxygen (Impact Sensitivity Threshold Technique), Test for

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia PA 19103.)

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(Industry association standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the requirements of this standard and the references cited herein, the requirements of this standard shall take precedence.

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3. DEFINITIONS

3.1 Background. The surface of the component against which an indication is viewed.

3.2 Black light. Electromagnetic radiation in the near-ultraviolet range. Wavelength is from 320 to 400 nm.

3.3 Black light filter. A filter that transmits near-ultraviolet radiation while absorbing other wavelengths.

3.4 Bleedout. The action of entrapped liquid penetrant in surfacing from discontinuities to form indications.

3.5 Component. The part(s) or element(s) of a system described, assembled, or processed to the extent specified by the drawing.

3.6 Contaminant. Any foreign substance present on the test surface or in the inspection materials which will adversely affect the performance of liquid penetrant materials.

3.7 Contracting agency. A prime contractor, subcontractor or Government agency procuring liquid penetrant inspection services.

3.8 Dark adaptation. The adjustment of the eye that enhances visual perception in a darkened area.

3.9 Defect. A discontinuity or group of discontinuities that exceed the specified acceptance criteria for a component.

3.10 Developer. A material applied to the test surface to accelerate bleedout and enhance the contrast of indications.

3.11 Developer, dry. A fine, free-flowing powder used as supplied.

3.12 Developer, nonaqueous. A developer that is either dissolved or suspended in a volatile solvent (other than water) as supplied, and rapidly dries to an adsorptive (or absorptive) coating.

3.13 Developer, wet (aqueous) soluble. A developer, normally supplied as a dry concentrate, that is completely soluble in water, and dries to an absorptive (or adsorptive) coating.

3.14 Developer, wet (aqueous) suspendible. A developer, normally supplied as a dry powder concentrate, that is suspended in water and dries to an adsorptive (or absorptive) coating.

3.15 Drain time. The portion of dwell time during which excess penetrant or emulsifier drains from the component.

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3.16 Drying oven. The oven used to increase the evaporation rate of rinse water or an aqueous developer from component.

3.17 Dwell time. The total time that a penetrant, emulsifier or remover, or a developer spends in contact with the component. For aqueous and nonaqueous developers, the dwell time starts after the developer is dry on the component.

3.18 Emulsifier. A liquid that renders the excess penetrant on the component surface water-washable.

3.19 Emulsifier, hydrophilic. An emulsifier that is water-soluble.

3.20 Emulsifier, lipophilic. An emulsifier that is oil-soluble and not water-soluble.

3.21 Etching. The uniform removal of surface material by chemical or electrochemical methods (also known as chemical milling).

3.22 Evaluation. The review following interpretation of indications that determines the acceptability of the component.

3.23 Fluorescence. The emission of visible radiation by a material as a result of, and only during, the absorption of black light.

3.24 Government procurement agency. The Government agency for which the system is built or service provided.

3.25 Hydro-air nozzle. Water nozzle that mixes compressed air with water.

3.26 Indication. The visible result of penetrant bleedout.

3.27 In-process. That which occurs during manufacturing before a component is in final form.

3.28 In-service. Refers to components that are in use or storage for their intended function.

3.29 Inspection. Visual examination under appropriate illumination of a component after completion of the liquid penetrant processing steps.

3.30 Interpretation. The determination of the cause(s) of indications and the classification of such cause(s).

3.31 Linear indication. Penetrant indications with at least a three to one length to width ratio.

3.32 NDT facility. The organization responsible for providing nondestructive inspection service.

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3.33 Penetrant. A solution of dye capable of entering discontinuities open to the surface.

3.34 Penetrant, post emulsifiable. A penetrant that requires the use of a separate emulsifier to render the penetrant water-washable.

3.35 Penetrant, solvent removable. A penetrant used in the process whereby the excess surface penetrant is removed with solvent. Postemulsifiable or water-washable penetrants can also be used in the solvent removable mode.

3.36 Penetrant, water-washable. A penetrant that contains an integral emulsifier or otherwise can be removed with water.

3.37 Postcleaning. The removal of residual penetrant inspection materials after evaluation.

3.38 Precleaning. The component cleaning done immediately before penetrant inspection processing.

3.39 Prime contractor. The organization that has responsibility to the Government for the total system. The prime contractor may be the Government in the procurement of such items as spare parts, special items, or basic materials.

3.40 Relevant indication. An indication of a true discontinuity.

3.41 Reprocess. Repeat, after cleaning, the application and appropriate processing of penetrant, emulsifier (as required) and developer (as required).

3.42 Solvent remover. A volatile solvent that is used to remove excess surface penetrant.

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4. GENERAL REQUIREMENTS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the prime contractor is responsible for the performance of all inspection requirements as specified herein. The prime contractor shall specify more stringent requirements than the minimum specified in this standard when necessary to assure that a component meets its functional and reliability requirements. Except as otherwise specified, the supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein. The Government reserves the right to perform any of the inspections set forth in the standard where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Specifying. When inspection is required in accordance with this standard, orders, contracts or other appropriate documents shall indicate the criteria by which components are judged acceptable. An example of such criteria is contained in MIL-P-47158; however, other criteria may be utilized. Engineering drawings or other applicable documents shall indicate the acceptance criteria for the entire component; zoning may be used. Inspection on a sampling basis shall not be allowed unless specifically permitted by the contract.

4.3 Personnel qualification. Personnel making accept/reject decisions described in this standard shall be qualified to at least Level II in liquid penetrant inspection in accordance with MIL-STD-410. Personnel performing the processing as described in this standard shall be qualified to at least Level I in liquid penetrant inspection in accordance with MIL-STD-410. If a Level I performs the processing, it shall be under the direct supervision or observation of a Level II or higher individual.

4.4 Materials.

4.4.1 Qualified materials. Only materials listed or approved for listing on QPL-25135 (reference MIL-I-25135) shall be utilized for penetrant inspection. Materials not conforming to the requirements of MIL-I-25135 may be used only when a waiver is obtained from the contracting agency.

4.4.2 LOX compatible materials. Penetrant materials tested in accordance with ASTM D2512 and passing at 70 foot-pounds (95J) or higher, shall be used on LOX (Liquid Oxygen) wetted surfaces which cannot be thoroughly post-cleaned. Use of these materials shall be in accordance with the material supplier instructions and shall require approval of the contracting agency when such materials do not meet the requirements of MIL-I-25135.

4.5 Equipment and facilities.

4.5.1 Processing. The penetrant inspection processing equipment shall be constructed and arranged to permit a uniform and controlled operation.

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The equipment shall meet all applicable national and local safety requirements as well as the requirements specified herein.

4.5.2 Viewing. Viewing areas shall be kept clean at all times. For visible dye inspection, Type II, the lighting system shall provide at least 100 foot-candles (1000 lx/m²)of white light at the surface of the component being inspected. For stationary fluorescent dye inspection, Type I, the ambient white light background shall not exceed 2 foot-candles (20 lx/m²) and the black lights shall provide a minimum of 800 micro watts/cm² when measured 15 inches (38 cm) from the front face of the black light filter or bulb. Viewing areas for portable fluorescent dye inspection shall utilize dark canvas, photographer's black cloth, or other methods to reduce the white light background to the lowest possible level during inspection and black light intensity shall be adequate.

4.6 Written procedure. All liquid penetrant inspections shall be performed to a specific written procedure that implements the requirements of this standard for the components under test. Since the inspection procedures are similar for many components, a master written procedure may be utilized that covers the details common to a variety of components. As a minimum, the following information is required either in individual procedures, a master procedure, or a combination thereof:

a. Details of the precleaning and etching process, including the materials used, the drying parameters and the processing times. If these operations are performed by other than inspection personnel, details concerning the operations may be specified in other documents but must be referenced in the procedure(s).

b. Classification of the penetrant inspection materials required in accordance with MIL-I-25135.

c. Complete processing parameters for the penetrant inspection materials including dwell times, application methods, drying times, concentrations, temperatures, and controls to prevent excessive drying of penetrant or overheating of component, as appropriate.

d. Complete inspection/evaluation requirements including light intensities (both inspection and ambient), the accept/reject criteria and the method and location of marking (see 5.9).

e. Identification of the components or areas within a component to be inspected in accordance with the procedure.

f. Complete postcleaning procedures. If postcleaning is performed by other than inspection personnel, details concerning this operation may be specified in other documents, but must be referenced in the procedure.

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4.7 Inspection sequence. Final penetrant inspection shall be performed after completion of all operations that could cause surface connected discontinuities. Such operations include, but are not limited to, grinding, welding, straightening, machining, and heat treating.

4.7.1 Surface treatment. Final penetrant inspection may be performed prior to treatments that can smear the surface but not by themselves cause surface discontinuities. Such treatments include, but are not limited to, vapor blasting, deburring, sanding, buffing, sandblasting, lapping, or peening. Performance of final penetrant inspection after such surface treatments requires that etching be included in the precleaning operation.

4.7.2 Surface coatings. Final penetrant inspection shall be performed before surface coatings such as paint, primer, anodize, plating, and thermal barrier, are applied. Final penetrant inspection may be performed after conversion coatings are applied provided that preparation for the coating included an etch. Chrome plated surfaces shall be inspected after final grinding only when specifically required. In-service components, unless otherwise specified, shall be inspected after removal of surface coatings and etching.

4.8 Materials and process limitations. Not all penetrant sensitivity levels, materials, and process methods are applicable to all inspection requirements. The sensitivity level shall be adequate for the intended purpose of the inspection. Unless there is an approval for deviation given by the contracting agency, the following selections are mandatory or forbidden, as indicated:

a. Forms a and b (dry powder and water soluble) developers shall not be used with Type II (visible dye) penetrant systems. This is not intended to prohibit the use of a Form e developer that has been qualified with a particular Type II system in accordance with MIL-I-25135.

b. Type II penetrant inspection shall not be used for final acceptance inspection of aerospace products. In addition, Type II penetrant inspection shall not be used prior to a Type I penetrant inspection of the same surface. This is not intended to eliminate the use of in-process Type II inspections where subsequent fabrication/forming operations remove the surfaces inspected.

c. The maintenance or overhaul inspection of turbine engine critical components shall be done only with Type I, Method D (post emulsified, hydrophilic) processes and either sensitivity Levels 3 or 4 penetrant materials.

d. Penetrant inspection without a developer shall be allowed only for those penetrant systems listed in QPL-25135 as having met the appropriate sensitivity level requirement without a developer. However, the use of a developer is mandatory for any in-service inspection.

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4.9 Records. The results of all penetrant inspections shall be recorded. All recorded results shall be identified, filed, and made available to the contracting agency upon request. Records shall provide for traceability to the specific part or lot inspected. As a minimum, the records shall include: a reference to the specific procedures used; location, classification, and disposition of relevant indications; the inspector's signature and certification level; and the date of inspection.

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5. SPECIFIC REQUIREMENTS

5.1 Surface Preparation. All surfaces to be inspected shall be clean, dry, and free of soils, oil, grease, paint and other coatings, corrosion products, scale, smeared metal, welding flux, chemical residues or any other material that could prevent the penetrant from entering discontinuities, suppress dye performance, or produce unacceptable background. Cleaning methods, including etching, selected for a particular component shall be consistent with the contaminants to be removed and shall not be detrimental to the component or its intended function.

5.1.1 Solvent cleaning. Solvent cleaning, which includes vapor degreasing and ultrasonic cleaning, shall be used for the removal of oils, greases, waxes and as the final cleaning step prior to penetrant inspection.

5.1.2 Chemical cleaning. Chemical cleaning shall be used for the removal of paints, varnishes, scale, carbon, or other contaminants that are not removable by solvent cleaning methods.

5.1.3 Mechanical cleaning. Mechanical cleaning shall be used for the removal of soils and other contaminants that cannot be removed by solvent or chemical cleaning methods.

5.1.4 Etching. Etching, unless otherwise specified, shall be performed when evidence exists that previous cleaning, surface treatments, or service usage has produced a surface condition that degrades the effectiveness of penetrant inspection. Etching processes shall be developed and controlled to prevent damage to the component under test. Etching is not required for those features such as close tolerance holes, close tolerance surfaces, faying surfaces, etc., where the function of the component or assembly would be degraded. Etching is not required for intermediate inspections when the surfaces(s) are not retained in the final part/component configuration or when the final penetrant inspection is preceded by etching.

5.2 Penetrant application and dwell. Unless otherwise specified, the entire surface of the component shall be covered with penetrant. Those areas not to be covered with penetrant shall be masked or otherwise protected. Penetrant shall be applied by spraying, dipping or brushing to provide coverage as required. The component, penetrant, and ambient temperatures shall all be in the range of 4° - 49°C (40° - 120°F) unless otherwise specified. The dwell time, unless otherwise specified, shall be a minimum of 10 minutes. Rotate or otherwise move components, if required, during dwell to prevent pooling of the penetrant. For dwell times greater than 2 hours, the penetrant shall be reapplied as required to prevent drying. The component shall be immersed in penetrant, if that is the application method, no more than 1/2 the total dwell time.

5.3 Penetrant removal.

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5.3.1 Method A process. Water-washable penetrants shall be removed with a manual or automated water spray or a manual wipe. Immersion in agitated water shall be permitted only for Level 1 and Level 2 sensitivity processes.

5.3.1.1 Manual spray. Maximum water pressure shall be 275 kPa (40 psi). Water temperature shall be between 10° - 38°C (50° - 100°F). A coarse spray shall be used with a minimum of 30 cm (12 inches), when possible between the spray nozzle and the part. Hydro-air nozzles shall be permitted only for Level 1 or Level 2 sensitivity processes with a maximum of 172 kPa (25 psi) added air pressure. Washing shall be conducted under appropriate illumination to assure that over-washing does not occur. Washing times shall be held to a minimum to prevent over-washing. If over-washing occurs, the component(s) shall be thoroughly dried and reprocessed. After rinsing, drain water from the component and utilize repositioning, suction, blotting with clean absorbent materials, or filtered shop air at less than 170 kPa (25 psi) to prevent pooling.

5.3.1.2 Automated spray. For automated spray systems, the wash parameters shall be such that the requirements of this standard are met. Water temperature shall be maintained between 10° - 38°C (50° - 100°F).

5.3.1.3 Manual wipe. Excess penetrant shall be removed with a clean, dry, lint free cloth or absorbent toweling. The remainder of the surface penetrant shall then be removed with a water dampened cloth or towel. The surface shall not be flushed with water and the cloth or towel shall not be saturated with water. The component shall be examined under appropriate illumination to assure adequate removal of the surface penetrant. Over-removal of the surface penetrant shall require that the component be cleaned and reprocessed. The surface shall be dried by blotting with a clean, dry towel or cloth, or by evaporation.

5.3.2 Method B process. Lipophilic post emulsified penetrant systems shall be removed by water immersion or with a water spray rinse after application of an emulsifier and an appropriate emulsifier dwell time.

5.3.2.1 Lipophilic emulsifier application and dwell. Lipophilic emulsifiers shall be applied by immersion or flowing. Lipophilic emulsifiers shall not be applied by spray or brush and shall not be agitated while on the surface of the component. Maximum dwell times, unless otherwise specified, shall be 3 minutes for Type I systems and 30 seconds for Type II systems. Actual dwell times shall be the minimum necessary to produce an acceptable background on the component.

5.3.2.2 Rinsing. After the appropriate emulsifier dwell time, emulsification shall be stopped by immersion or water spray. For spray removal of the penetrant/emulsifier mixture, the parameters of 5.3.1 apply. Dwell time in an agitated immersion rinse, if used, shall be the minimum required to remove the emulsified penetrant. Examine the components under appropriate illumination after rinsing. Clean and reprocess those components with

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excessive background. The water used in rinsing shall be free of any contaminant that will leave a residue on the component when dried. After rinsing, drain water from the component and utilize repositioning, suction, blotting with clean absorbent materials or filtered shop air at less than 170 kPa (25 psi) to prevent pooling. If over-emulsification is observed, the component must be cleaned and reprocessed.

5.3.3 Method C process. Solvent removable penetrants are removed by first wiping the excess penetrant with a clean, lint free, dry cloth or absorbent toweling. The remainder of the surface penetrant is then removed with a solvent dampened lint free cloth or towel. The surface of the component shall not be flushed with solvent and the cloth or towel shall not be saturated with solvent. The component and cloth or toweling shall be observed under appropriate illumination to assure adequate removal of the surface penetrant. Over-removal of the surface penetrant shall require the component to be cleaned and reprocessed. The surface shall be dried by blotting with a lint free dry cloth or towel, or by evaporation.

5.3.4 Method D process. Hydrophilic post emulsified penetrant systems shall be removed with a water prerinse, application of the hydrophilic emulsifier and then a postrinse.

5.3.4.1 Prerinse. The water prerinse shall be applied for the minimum amount of time required to achieve removal of the bulk surface penetrant. The rinse parameters of 5.3.1 shall apply.

5.3.4.2 Hydrophilic emulsifier application and dwell. The hydrophilic emulsifier shall be applied by immersion, flowing, foaming, or spray. For immersion applications, the concentration shall be no higher than specified by the penetrant system supplier and shall not exceed 35 percent by volume. For immersion applications, the emulsifier or part shall be mildly agitated. Dwell time shall be the minimum required for adequate penetrant removal, but unless otherwise specified, shall not exceed 2 minutes. For spray applications, the concentration shall not exceed 5 percent.

5.3.4.3 Postrinse. After the application and dwell of the hydrophilic emulsifier, the component under test shall be rinsed with water. The water shall be free of contaminants that could leave a residue on the component after drying. The spray rinse parameters of 5.3.1 shall apply. Evidence of over-removal shall require the part to be cleaned and reprocessed. Excessive background may be removed by additional (touch-up) application of the hydrophilic emulsifier provided the maximum allowable dwell time is not exceeded for the hydrophilic emulsifier. Additional rinsing of the touch-up area will be required after application and dwell of the hydrophilic emulsifier. If careful touch-up application of the hydrophilic emulsifier does not produce an acceptable background, the part shall be cleaned and reprocessed. Manual systems shall require the use of appropriate illumination to assure adequate penetrant removal.

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5.4 Drying.

5.4.1 Sequence. The components shall be dried prior to the application of dry developer, nonaqueous developer, or inspection without developer. The components may be dried before the application of aqueous soluble or suspendible developers. The components shall be dried after the application of aqueous developers.

5.4.2 Drying parameters. Components shall be air dried at room temperature or in a drying oven. Oven temperature shall not exceed 70°C (160°F) and drying time shall not exceed that necessary to adequately dry the component. Components shall not be placed in the oven with pooled water or pooled aqueous solutions/suspensions.

5.5 Developing. Unless otherwise specified, developers shall be utilized for penetrant inspection in accordance with the requirements of this standard. Type I penetrants that are qualified to MIL-I-25135 without the standard dry developer are so noted in QPL-25135 and may be used without developer. Minimum and maximum dwell times without developer shall be 10 minutes and 2 hours respectively. For all developing conditions, components that are not inspected before the maximum dwell time shall be cleaned and reprocessed.

5.5.1 Dry developers. Components shall be dry before the developer is applied. Dry developer shall be applied in such a manner as to contact all surfaces to be inspected. Minimum dry developer dwell shall be 10 minutes; the maximum dwell shall be 4 hours. Excess dry developer may be removed after the dwell time by light tapping. Dry developers shall not be used with Type II penetrants.

5.5.2 Nonaqueous developers. Components shall be dry before application of the developer. Nonaqueous developer shall be applied by spraying only. For Type I penetrants the developer shall be applied as a uniform thin coating over the entire surface to be inspected. For Type II penetrants, the developer shall be applied over the entire surface to form a uniform, white coating to provide suitable color contrast for the penetrant indications. The uniformity and thickness of the developer coating is important for both types of penetrant systems. If the developer coating thickness is too heavy for Type I systems such that the metallic surface is completely masked, the component shall be cleaned and reprocessed. The minimum and maximum dwell times for nonaqueous developers are 10 minutes and 1 hour respectively. For nonaqueous suspendible developer, the developer container shall be frequently agitated during application.

5.5.3 Aqueous developers, soluble and suspendible. Aqueous developers may be applied to the component after rinsing, or they may be applied after the component is dry. Aqueous soluble developers shall not be used with Type II penetrants or Type I, Method A penetrants unless otherwise specified. Aqueous suspendible developers, unless otherwise specified, shall not be

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used with Type I penetrants. Developers shall be applied by spray, flowing or immersion. The applied developer shall not be allowed to puddle and shall completely cover all surfaces to be inspected. Components shall be air dried or oven dried to the requirements of 5.4.2. Minimum and maximum dwell times, after the component is dry, are 10 minutes and 2 hours.

5.6 Inspection. The viewing area shall meet the appropriate requirements of 4.5.2. Components shall be inspected before the maximum developing dwell time, and if required by specific procedures, monitored periodically during the developing dwell time. Components not inspected before the maximum developing dwell time shall be cleaned and reprocessed.

5.6.1 Type I processes. Personnel shall be dark adapted for a minimum of 1 minute prior to inspecting components and shall not wear photochromic or permanently darkened lenses. Black lights shall provide a minimum of 1200 micro watts/cm^2 at the component surface and be positioned so that the inspection of the component is not impeded. All areas of fluorescence shall be interpreted. Components with no indications or only non-relevant indications shall be accepted. Components with relevant indications shall be evaluated against the appropriate acceptance criteria. Components with excessive background fluorescence shall be cleaned and reprocessed.

5.6.2 Type II processes. All indications shall be interpreted. Components with no indications or only non-relevant indications shall be accepted. Components with relevant indications shall be evaluated against the appropriate acceptance criteria. Components with excessive background shall be cleaned and reprocessed.

5.6.3 Evaluation. If allowed by the specific procedure, indications may be evaluated by wiping the indication with a solvent dampened swab, allowing the area to dry, and redeveloping. Redevelopment time shall be as long as the original development time. If no indication re-appears, the original indication is considered false. This procedure may be performed twice for any given original indication. Unless allowed by a specific procedure, indications of cracks, laps, and folds shall be cause for rejection. Indications that exceed the specific acceptance criteria for the component shall be cause for rejection.

5.6.3.1 Mechanical evaluation. When allowed by the specific inspection procedure, indications may be removed by an approved procedure such as sanding, either powered or manual, or grinding to determine the depth and extent of the discontinuity. After the mechanical operation, the area shall be cleaned, etched, and re-inspected. The process used for re-inspection shall be at least as sensitive as the original process.

5.7 Postcleaning. Components shall be cleaned after inspection to remove developers and other inspection material residues if they are detrimental to subsequent operations or the components' intended function.

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5.8 Quality assurance provisions. This section provides the controls necessary to assure that the penetrant system materials and equipment provide an acceptable level of performance. The frequency of the required checks is based upon a facility operating for a full, one-shift-operation daily. For facilities operating less frequently, the frequency of daily and weekly checks may be reduced, but must be performed prior to inspections. Other checks should be performed at the same frequency as for full time operations. The NDT facility may perform these process control operations or contract for their performance with an independent laboratory.

5.8.1 Black lights. All black lights shall be checked daily, and after bulb replacement, for output. A longer period may be used if a plan justifying this extension is prepared by the NDT facility and approved by the contracting agency. Minimum acceptable intensity is 800 micro watts/cm^2 measured at a distance of at least 38 cm (15 inches) from the front of the bulb or filter. Black light reflectors and filters shall be checked daily for cleanliness and integrity. Damaged or dirty reflectors or filters shall be replaced or corrected as appropriate.

5.8.2 White light. Viewing areas for Type II penetrant systems shall be checked weekly to assure a minimum of 100 foot-candles (1000 lx/m^2) white light intensity at the working level (for example, bench or table top).

5.8.3 System performance. The penetrant system shall be checked daily for performance. The check shall be made with known defect standards. The check shall be performed by processing the known defect standard through the system using appropriate processing parameters and comparing the indications thus obtained to those obtained with unused samples of the same materials. This comparison may be made with photographs or other such records of previously obtained indications or with a similar known defect standard processed with the unused materials. When the performance of the in-use materials falls below the performance of the unused materials, the in-use material quality shall be checked with the appropriate tests and corrected prior to conducting any penetrant inspection in accordance with this standard.

5.8.3.1 Known defect standards. The selection and procedures for the maintenance of known defect standards shall be approved by the contracting agency. The defects in the standard shall be capable of demonstrating unsatisfactory system performance. The maintenance procedures shall assure that cleaning of the standards between usages is adequate and that physical changes in the standard that make it unsuitable for use can be detected.

5.8.4 Materials. Prior to being placed in use, the conformance of materials to the requirements of MIL-I-25135 shall be verified. Use of materials not conforming to MIL-I-25135 shall be approved by the contracting agency and shall be allowed only when materials conforming to MIL-I-25135 are inadequate for the particular application. Operators shall be alert to any changes in performance, color, odor, consistency or appearance of all

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penetrant materials in use and shall conduct the appropriate checks and tests if they have reason to believe the quality may have deteriorated. Penetrant inspection shall be conducted in accordance with this standard only after acceptable quality of materials has been established.

5.8.4.1 Penetrants. The following checks on in-use penetrants shall be conducted at least monthly with the exception of brightness which shall be conducted at least quarterly. Unsatisfactory performance shall be cause for replacement of the in-use material or corrective action as appropriate.

5.8.4.1.1 Brightness. Brightness tests of in-use penetrants shall be conducted in accordance with MIL-I-25135 with a sample of the unused penetrant serving as the reference. Brightness values less than 90 percent of the unused penetrant brightness are unsatisfactory.

5.8.4.1.2 Water content (Method A penetrants only). Water content of in-use Method A penetrants shall be measured in accordance with ASTM D 95. Water content in excess of 5 percent by volume is unsatisfactory.

5.8.4.1.3 Removability (Method A penetrants only). The removability of Method A penetrants shall be tested in accordance with MIL-I-25135 with a sample of the unused penetrant serving as a reference. If the removability is noticeably less than the reference, the in-use penetrant shall be replaced.

5.8.4.1.4 Sensitivity. The sensitivity of penetrants shall be checked in accordance with the procedures of 5.8.3 where the in-use penetrant is used with the unused emulsifier, if applicable, and unused developer if applicable and compared to the results obtained using the unused penetrant, unused emulsifier, if applicable, and unused developer. Sensitivity of the in-use penetrant noticeably less than the reference is unsatisfactory.

5.8.4.2 Emulsifiers. The following checks on in-use emulsifiers shall be performed as specified. Unsatisfactory performance shall be cause for replacement of the in-use emulsifier or corrective action as appropriate.

5.8.4.2.1 Removability. Removability of the in-use emulsifier shall be checked weekly in accordance with MIL-I-25135. The in-use emulsifier will be used with the unused penetrant and compared to the reference system of unused emulsifier used with the unused penetrant. Removability less than that of the reference system is unsatisfactory.

5.8.4.2.2 Water content (lipophilic emulsifier). Water content shall be checked monthly in accordance with ASTM D 95. Water content in excess of 5 percent is unsatisfactory.

5.8.4.2.3 Concentration (hydrophilic emulsifier). Concentration of immersion solutions shall be checked weekly with a refractometer. Variation of concentration greater than three percentage points from the initial,

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unused concentration is unsatisfactory. Spray concentrations shall be checked weekly with the refractometer and shall not exceed the specified concentration for the system.

5.8.4.3 Developers. The following checks on in-use developers shall be performed as specified. Unsatisfactory performance shall be cause for replacement or corrective action as appropriate.

5.8.4.3.1 Dry developer. Dry developer shall be checked daily to ensure it is fluffy and not caked. Caked dry developer is unsatisfactory. For dry developer that is recycled, 10 or more fluorescent specks observed in a 10 cm (4 inch) circle, when a sample is spread into a thin layer on a flat surface, is unsatisfactory.

5.8.4.3.2 Aqueous developer (soluble and suspended). Aqueous developers shall be checked daily for fluorescence, as appropriate, and coverage. Immerse a clean aluminum panel, about 8 cm x 25 cm (3 inches x 10 inches) and remove for drying and observation under a black light, as appropriate. Failure to uniformly wet the panel or observed fluorescence is unsatisfactory. Concentration shall be checked weekly with a hydrometer and shall be in accordance with the developer supplier recommended concentration.

5.8.5 Inspection area. The inspection area for stationary Type I systems shall be checked weekly for cleanliness, freedom from fluorescent contamination, and residual white light background. Background white light shall not exceed 2 foot-candles (20 lx/m^2), and the area shall be free of clutter and fluorescent contamination.

5.8.6 Operating pressures/temperatures. Indicators and controls shall be checked at the start of each shift to assure proper settings. Those indicators displaying out of control settings shall be adjusted to the proper settings. Indicators and controls shall be calibrated at intervals in accordance with MIL-STD-45662.

5.9 Marking. Unless otherwise specified, each component which has been penetrant inspected and found to conform to the requirements of this standard and the acceptance requirements of the contract or purchase order shall be marked as specified herein. Marking shall be applied in a manner and location that is harmless to the component, or its intended function, and to preclude removal, smearing or obliteration by subsequent handling. When subsequent processing would remove such identification, the records accompanying the component shall be marked. The methods of marking are listed below in descending order of preference. Marking shall conform to MIL-STD-792.

5.9.1 Impression stamping. Impression stamping shall be used where explicitly permitted by applicable specifications or drawings. Marking shall be located in areas adjacent to the part number or the inspector's stamp.

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5.9.2 Etching. If impression stamping is not permitted, parts may be marked by etching, and suitable etchants and application methods shall be used. Etching methods other than chemical or electrochemical may be used.

5.9.3 Dyeing. Where etching or impression stamping is not permitted, identification may be accomplished by dyeing or ink stamping.

5.9.4 Other identification. Other means of identification, such as tagging, may be applied when the construction, finish, or functional requirements of the component preclude etching, dyeing, or stamping. Items such as bolts or nuts may be identified by conspicuously marking each package.

5.9.5 Symbols. Each component that has successfully passed inspection shall be marked as follows:

a. When etching or stamping is applicable, symbols shall be used. The stamping may contain an identification symbol of the facility and the last two digits of the year of inspection.

(1) Except for specialized applications, use the symbol P to denote 100 percent inspection.

(2) All components, in the lot sampled, accepted on sampling basis shall be marked with the symbol P enclosed by an ellipse.

b. When dyeing is used, maroon dye shall be used to denote components accepted on a 100 percent inspection basis. Yellow dye shall be used to denote a sampling basis.

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6. NOTES

6.1 Intended use. The penetrant inspection process is intended for the detection of discontinuities open to the surface, such as lack of fusion, corrosion, cracks, laps, cold shuts and porosity.

6.2 Nonmetallic components. Some cleaners, penetrants, and developers can have a deleterious effect on nonmetallic materials such as plastics. Prior to inspection, tests should be conducted to ensure that any cleaning or inspection materials are not harmful to the components to be examined. Caution must also be exercised in the usage of elevated temperature with components manufactured from thermoplastic materials.

Custodians:
Air Force - 20
Navy - AS
Army - MR

Reviewers:
Air Force - 68, 82, 99
Navy - OS, SH
Army - AR, EA
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Preparing Activity:
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